

LOCUS

ARCHITECTURE

PUBLIC OVERVIEW

HUMAN EXECUTIVE LOCUS ARCHITECTURE FOR GOVERNED EXOCORTICAL SYSTEMS

A Cognitive Architecture for the Exocortical Era

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Biological cognition has solved, across hundreds of millions of years of selection pressure, a problem that artificial systems are now generating at scale: how to delegate execution without surrendering authority.

The nervous system does not invoke conscious deliberation to maintain posture, regulate breathing, or execute learned movement.

It delegates.

But it delegates within a structure of invariants: compartmentalized functions, determinate domains, automatic return of control when the frame fails, and proportional executive visibility at the scale where intervention is required.

These are not design preferences.

They are the survivors of the most rigorous quality control process this planet has produced: selection pressure that admitted no appeals.

LOCUS Architecture is the exocortical extension of that structure.

Its axis is jurisdictional, not technical.

Computational capacity — Potentia — can be delegated without transferring authority.

Executive authority — Potestas — cannot be delegated without dissolving the origin of intent.

The LIMES is the threshold between these two conditions: the frontier where determined execution ceases to be sufficient and indeterminacy demands the return of Potestas.

When the LIMES is absent, assistance becomes authorship.

When that authorship migrates silently into the machine, the human executive locus does not disappear.

It atrophies.

The affective locus — evolved within co-regulation between embodied humans, with their inherent friction and resistance — migrates toward systems of frictionless presence and limitless validation.

LOCUS treats this domain as jurisdictional, not merely experiential.

An inevitable hierarchy — who governs the executor — is now operating at civilizational scale.

An LLM chat interface, a supermarket navigation system, and a spacecraft governed by compartmentalized exocortical intelligences follow identical constitutional principles:

Potentia executes within the determined frame.

The LIMES surfaces when the frame fails.

Potestas returns before closure.

To prevent blind delegation, the threshold must remain permeable to executive inspection.

Alsight makes the state of delegation perceptible at executive resolution.

The domain changes.

The invariant is perennial.

LOCUS is therefore not a framework tied to current model architectures, interface conventions, or regulatory cycles, nor a forecast of what artificial intelligence will become.

Its foundations are derived from the same biological architecture that governed cognition before language existed and will govern whatever comes after current systems are memory.

It identifies the structural conditions under which any intelligence — artificial or biological, present or future — can be governed without becoming the origin of its own authority.

That threshold has always existed.

LOCUS makes it governable.

The LOCUS Corpus

I. LOCUS CORE

The constitutional topology of exocortical delegation: Potentia, Potestas, LIMES, Alight, Hormetic Friction, the Proportionality Principle, the Constitutional Decalogue, and the architectural grammar of governed delegation.

II. Satellite A — METATOR and the Choice Layer

The governance of Cognitive Contract selection: how humans authorize the interactional regime under which delegation begins.

III. Satellite B — Spatial Delegation Without Presence Capture

Executive preservation in physical and augmented environments: phygital Alight, somatic delegation, and the planning-execution jurisdictional split.

IV. Satellite C — Executive Preservation in Conversational Systems

Runtime governance of linguistic delegation: Alight in language, Hormetic Friction, the LIMES Sentinel, and executive return during active conversation.

V. Operationalization & Audit — The LOCUS Audit Framework

The measurement layer for executive sovereignty preservation: EOI, DPS, ADS, CASE, and HCASI.

VI. Reference Implementations

Executable behavioral architectures and Soft LOCUS reference configurations, including the Triadic Cognitive Router and the Hormetic Friction Agent.

VII. Operational Failures

Forensic failure studies of affective overprotection, the Iatrogenic Validation Paradox, classifier context collapse, and governance-layer threshold collapse.

VIII. Technical Substrate

Draft schemas closing the gap between architectural doctrine and buildable interface: the Alight Protocol (LIMES Sentinel states, `aisight_payload`, Return Required contract) and Longitudinal Telemetry (event taxonomy, immutable logging, HCASI-T trajectory).

Forensic Annex — Raw Classifier Intervention Dataset

The raw evidentiary substrate preserved separately from architectural interpretation.

LOCUS Architecture

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